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CHEM 237: Organic Chemistry I

Institution: Pacific Crest University (fictional) **Credits:** 4 (quarter hours) **Instructor:** Staff
Prerequisite: CHEM 142 and CHEM 152 **Contact hours:** 4 lecture hours per week, 10-week quarter. Laboratory is a separate course (CHEM 241).

Course description

Organic chemistry I (quarter): bonding, structure, stereochemistry and an introduction to substitution reactions. Elimination reactions and alkenes are covered in the next quarter. Laboratory is a separate course.

Learning outcomes

- Explain bonding, hybridization and resonance in organic molecules
- Name organic compounds using IUPAC rules
- Analyse conformations and stereochemistry, including chirality
- Predict SN1 and SN2 substitution mechanisms

Weekly schedule

Week	Topic
1	Structure and bonding; hybridization
2	Resonance and acid–base chemistry
3	Functional groups and IUPAC nomenclature
4	Alkanes and conformational analysis
5	Midterm; cycloalkanes
6	Stereochemistry and chirality
7	Stereoisomers and optical activity
8	Nucleophilic substitution: SN2
9	Nucleophilic substitution: SN1
10	Review; final exam

Assessment

Problem sets 20%, two midterms 40%, final 40%. Elimination reactions, alkenes and alkynes are covered in CHEM 238.

Textbook: instructor-provided open course notes.

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